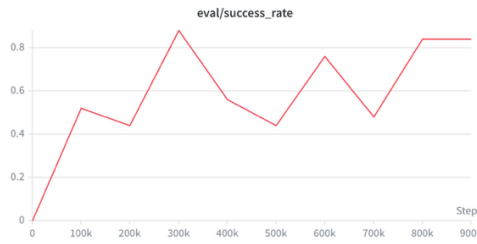


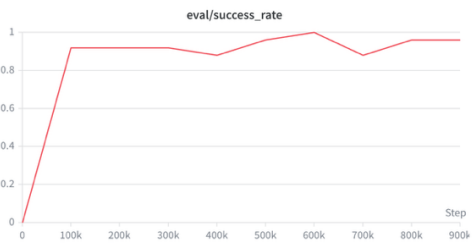
1. SAC + BC

a) Training curves of *cube-single-play-singletask-task1-v0*:

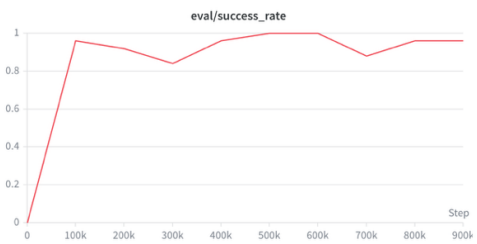
$\alpha = 30$



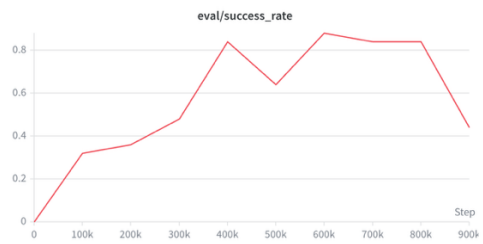
$\alpha = 100$



$\alpha = 300$



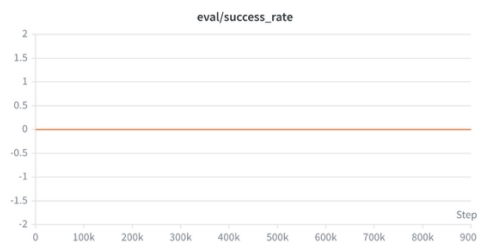
$\alpha = 1000$



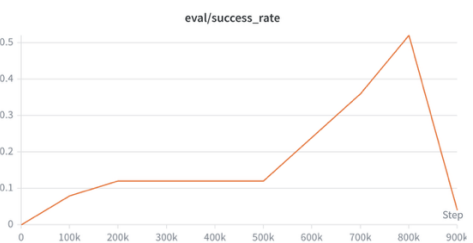
On the cube-single task, SAC+BC shows a clear dependence on the BC coefficient α . Moderate values, especially $\alpha = 100$ and $\alpha = 300$, achieve the best overall performance. In contrast, $\alpha = 30$ still produces a successful policy but with noticeably larger fluctuations, as weaker behavioral regularization introduces more OOD bias and variance. When α is increased to 1000, performance becomes significantly worse and eventually drops late in training, as overly strong BC regularization hinders policy exploration on given dataset.

b) Training curves of *antsoccer-arena-navigate-singletask-task1-v0*:

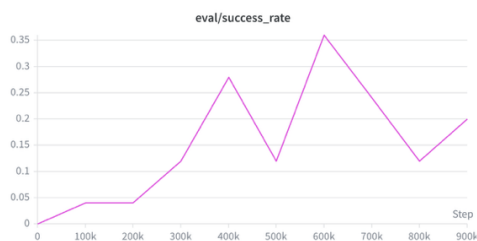
$\alpha = 1$



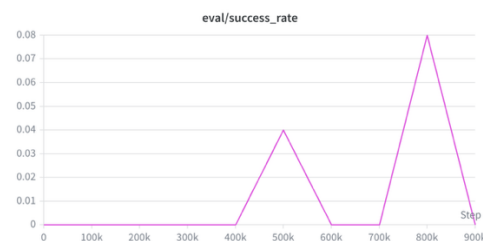
$\alpha = 3$



$\alpha = 10$



$\alpha = 30$

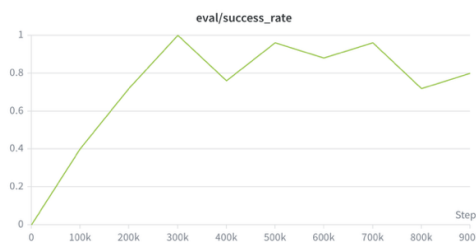


The same conclusion holds for this experiment as well: Moderate values, especially $\alpha = 3$, achieve the best overall performance, and $\alpha = 10$ also exceeds the target with weaker stability. In contrast, $\alpha = 1$ fails to learn useful behavior, while $\alpha = 30$ becomes too conservative and only briefly surpasses the threshold, again suggesting that overly strong behavioral regularization can degrade policy improvement.

2. IQL

a) Training curves of *cube-single-play-singletask-task1-v0*:

Alpha = 1



Alpha = 3



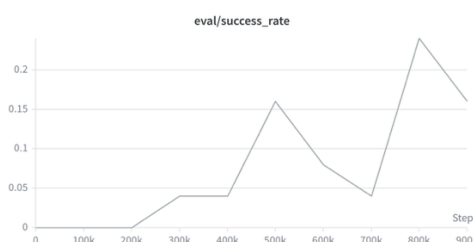
Alpha = 10



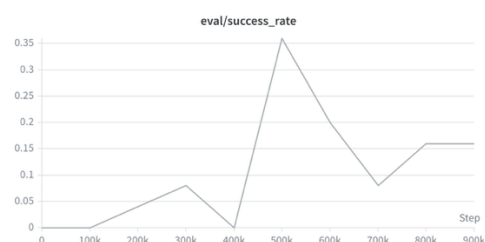
On the cube-single task, IQL performs well across all three tested values of alpha, and all of them exceed the required success rate. This suggests that IQL is relatively insensitive to alpha on this task. Among the three settings, alpha = 1 gives the best overall performance, while alpha = 10 still works but shows slightly larger fluctuations and a lower final success rate.

b) Training curves of *antsoccer-arena-navigate-singletask-task1-v0*:

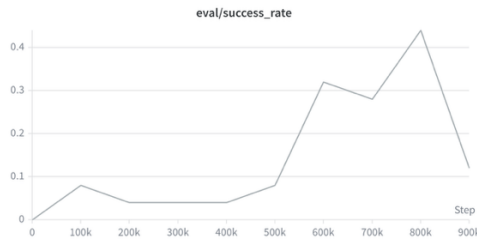
Alpha = 1



Alpha = 3



Alpha = 10



On antsoccer-arena, the training curves are noticeably noisier than on cube-single due to the higher difficulty of the task. Among the tested values, $\alpha=3$ provides the best balance between stability and peak performance, while $\alpha=10$ reaches the highest peak success rate but is less stable overall.

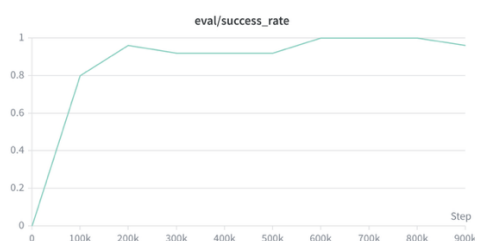
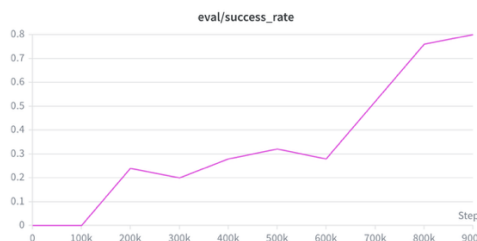
- c) SAC+BC achieves slightly better peak performance than IQL on some tasks, especially on cube-single, when the BC coefficient is carefully tuned. However, IQL is noticeably less sensitive to the choice of α , as a wide range of values can achieve good performance, whereas SAC+BC performance varies much more dramatically across different α values.

3. FQL

- a) Training curves of *cube-single-play-singletask-task1-v0*:

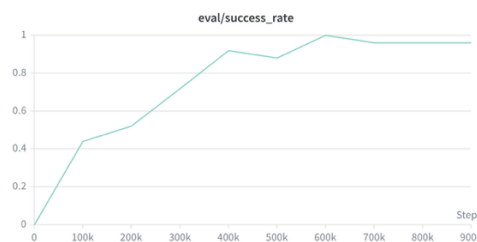
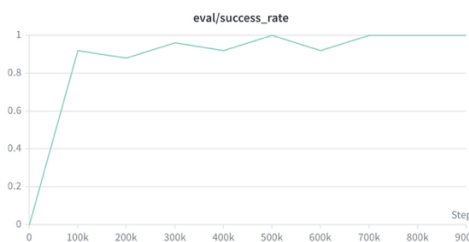
Alpha = 30

Alpha = 100



Alpha = 300

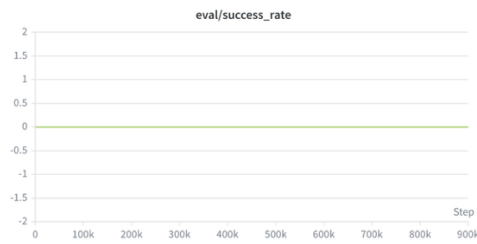
Alpha = 1000



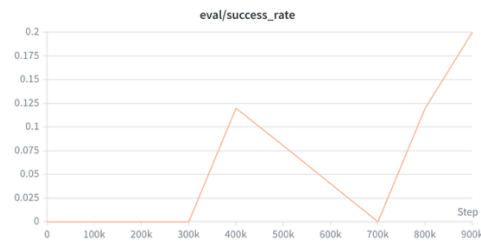
Like SAC+BC, FQL performs best with moderate BC coefficients such as $\alpha = 100$ and $\alpha = 300$. Smaller values lead to weaker regularization and lower final performance, while overly large values slow down learning because the policy stays too close to the behavior policy.

b) Training curves of *antsoccer-arena-navigate-singletask-task1-v0*:

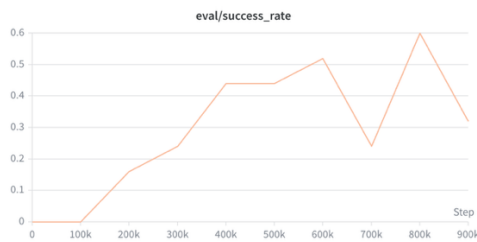
$\text{Alpha} = 1$



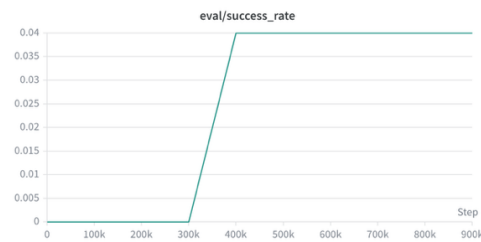
$\text{Alpha} = 3$



$\text{Alpha} = 10$



$\text{Alpha} = 30$



On AntSoccer, FQL is highly sensitive to the BC coefficient. Very small values lead to severe OOD issues, while overly large values over-constrain the policy. A moderate value such as $\alpha = 10$ gives the best trade-off and achieves the highest success rate.